

Shivansh Khullar

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EDUCATION

University of Waterloo

Waterloo, ON

Candidate for Bachelor of Applied Science in Computer Engineering

TECHNICAL SKILLS

Programming Languages: Python, C++, Java, JavaScript, TypeScript, SQL, Ruby

ML/AI: PyTorch, TensorFlow, Hugging Face Transformers, OpenCV, scikit-learn, YOLOv8, MediaPipe, LLM Fine-Tuning

Data & Pipelines: Pandas, NumPy, ETL Pipelines, BeautifulSoup, Matplotlib

Infrastructure & Tools: AWS, GCP, Docker, Git/GitHub, PostgreSQL, MongoDB, GraphQL, REST APIs, CI/CD

EXPERIENCE

AI Software Engineering Intern

May 2026 – Present

BaseLeaf Technologies Inc.

Waterloo, ON

- Engineered an **automated outreach pipeline** leveraging **Python**, **REST APIs**, and **event-driven webhooks**, automating **500+** monthly client touchpoints and reducing manual outreach effort by **85%**.
- Fine-tuned **transformer-based LLMs** and optimized inference pipelines for automated immigration case report generation, reducing report turnaround time from **45** to under **5 minutes** while improving output consistency by **35%**.
- Designed **automated QA workflows** for LLM-generated reports, combining **custom validation metrics** and **human-in-the-loop feedback**, improving factual consistency by **32%** across production outputs.

Software Engineer

Aug. 2025 – Jan. 2026

Stealth Startup

Toronto, ON

- Improved CRM data accessibility by **35%** through secure **Node.js**, **TypeScript**, and **GraphQL** API integrations, unifying data pipelines across multiple dealership systems.
- Cut duplicate records by **40%** and reduced preprocessing time from hours to minutes by optimizing **PostgreSQL** schema design and developing automated **XLSX validation scripts**.
- Deployed scalable **AWS Lambda** backend services with **RBAC** and logging, enabling real-time KPI tracking for a live **Ford** dealership pilot, with infrastructure designed to scale to **50+** dealerships with planned rollout to **Audi** dealerships.

Autonomy Software Developer

Sept. 2025 – Dec. 2025

Waterloo Aerial Robotics Group

Waterloo, ON

- Enhanced object detection accuracy by up to **20%** by researching and fine-tuning computer vision models using **Python**, **OpenCV**, and **ROS** for autonomous navigation in **SITL** simulations.
- Accelerated testing by **30%** through developing **sensor and GPS emulation modules**, improving the reliability of simulation-based validation for autonomous algorithms.

PROJECTS

Brain2Qwerty-on-EMG | Python, PyTorch, CTC, Transformers, Kaggle

July 2026

- Reproduced and benchmarked a **CTC encoder-decoder** pipeline on the **emg2qwerty** surface-EMG dataset (**30** users, **366** sessions, **111** hours), comparing an **unconstrained LLM correction** approach against a **constrained beam-search n-gram LM**, reducing **Character Error Rate** from **46.4%** to **24.8%**.
- Diagnosed a **signal-fidelity bottleneck** by re-processing raw **2kHz** EMG data instead of a **4x-decimated** 500Hz version, cutting mean CER by **14.5** points with **no architecture changes**, demonstrating that **unconstrained LLM correction degrades** noisy-signal decoding versus constrained re-ranking.

Sonata - Real-Time Gesture Music Interface | Python, OpenCV, NumPy, MediaPipe, Pygame

Jan. 2026

- Built a **gesture-controlled musical instrument** using real-time **hand-landmark tracking** to map **5+** distinct gestures to instrument sounds, enabling touchless, interactive music creation via webcam input.
- Processed **30+** FPS video streams with optimized computer-vision pipelines to ensure **low-latency audio feedback**, supporting smooth, continuous play during live gesture-based performances.

Sign Language Recognition System | Python, MediaPipe, OpenCV, scikit-learn, NumPy, pytsx3

Dec. 2025

- Developed a **multi-language gesture recognition system** converting **ASL**, **BSL**, **SSL**, and **ISL** gestures into text and speech, processing webcam streams to enable real-time communication for visually and speech-impaired users.
- Trained and validated ML classifiers on **4+** sign language datasets with **5,000+** images each, achieving **90%+** average accuracy and enabling seamless gesture-to-speech/text translation across multiple languages.

PredictPL | Python, Pandas, NumPy, Scikit-learn, Matplotlib, BeautifulSoup

Sept. 2025

- Designed an end-to-end **ML pipeline** analyzing **1,300+** Premier League matches with rolling-average feature engineering, training ensemble models to **~61%** accuracy and visualizing results through interactive dashboards.

CERTIFICATIONS

Machine Learning Engineer (**Google**), Advanced Generative AI for Developers (**Google**), Cloud Practitioner Essentials (**AWS**), CS50's Introduction to Programming with Python (**Harvard University**)